INCH-POUND

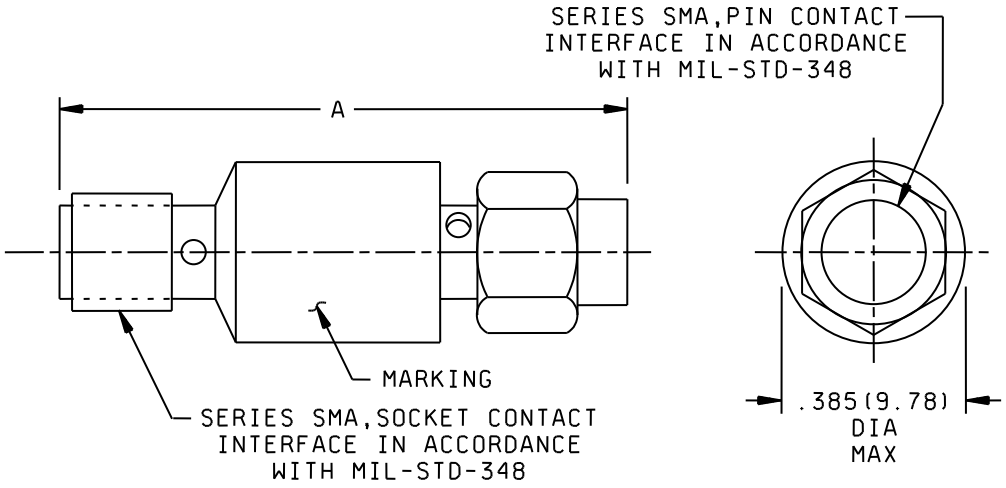
MIL-DTL-3933/16E
w/AMENDMENT 1
12 April 2021
SUPERSEDING
MIL-DTL-3933/16E
7 April 2011

DETAIL SPECIFICATION SHEET

ATTENUATORS, FIXED, SPACE LEVEL, NON SPACE LEVEL COAXIAL LINE (SMA CONNECTORS), FREQUENCY RANGE: DC TO 18 GHz, CLASS IV, LOW POWER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the attenuators described herein shall consist of this specification sheet and MIL-DTL-3933.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Metric equivalents are in parentheses.
4. Body diameter dimension requirement includes both the diameter of the component body and the Thickness of any externally attached marking label(s).
5. Widths across coupling nut flats are to accommodate wrench, normal size of .3125 (7.938) minimum in accordance with FED-STD-H28, appendix 10, wrench openings.
6. Width of coupling nut hex shall be .150 (3.81) minimum.

FIGURE 1. Dimensions and configuration.

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ENGINEERING DATA:

Operating frequency range: dc to 18 GHz.

Test frequencies for VSWR and attenuation measurements: dc, 4, 8, 12.4, and 18 GHz.

Basic test frequency: 8.0 GHz.

Operating temperature range: -55°C to +125°C.

Maximum weight: See table I.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Nominal impedance: 50 ohms.

Attenuation: See table I.

Power: See table I.

VSWR: See table I.

Power sensitivity: 0.005 dB/dB/watt maximum.

Temperature sensitivity: 0.0006 $\frac{dB}{dB \quad x \quad °C}$ maximum.

RF leakage: 80 dB below the input power level over the frequency range of dc to 18.0 GHz.

Part or Identifying Number (PIN): M3933/16- (and dash number from table I with three device level designators are space level (T), screened (S) and non-screened (N) options).

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TABLE I. Electrical characteristics.

Dash number	Attenuation dB		Maximum power input @ 25°C		VSWR maximum 1/				Weight (oz) Max	Dimension A (inches) maximum
	Nominal	Deviation DC to 18 GHz	AV (W) 3/ (continuously)	Peak 2/ (kW)	dc to 4.0 GHz	4.0 to 8.0 GHz	8.0 to 12.4 GHz	12.4 to 18.0 GHz		
01 NST	3	± 0.3	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
02 NST	6	± 0.3	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
03 NST	10	± 0.5	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
04 NST	20	± 0.7	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
05 NST	1	± 0.4	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
06 NST	2	± 0.4	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
07 NST	4	± 0.3	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
08 NST	5	± 0.3	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
09 NST	7	± 0.5	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
10 NST	8	± 0.5	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
11 NST	9	± 0.5	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
12 NST	30	± 1.0	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.8	1.850 (46.99)
13 NST	40	± 1.0	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.8	1.850 (46.99)
14 NST	50	± 2.0	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.8	1.850 (46.99)
15 NST	60	± 2.0	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.8	1.850 (46.99)
16 NST	0	± 0.4	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)

See footnotes at end of table.

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TABLE I. Electrical characteristics - Continued.

Dash number	Attenuation dB		Maximum power input @ 25°C		VSWR maximum 1/				Weight (oz) Max	Dimension A (inches) maximum
	Nominal	Deviation DC to 18 GHz	AV (W) 3/ (continuously)	Peak 2/ (kW)	dc to 4.0 GHz	4.0 to 8.0 GHz	8.0 to 12.4 GHz	12.4 to 18.0 GHz		
17 NST	0.5	± 0.4	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
18 NST	1.5	± 0.4	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
19 NST	2.5	± 0.4	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
20 NST	3.5	± 0.3	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
21 NST	4.5	± 0.3	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
22 NST	5.5	± 0.3	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
23 NST	6.5	± 0.3	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
24 NST	7.5	± 0.5	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
25 NST	8.5	± 0.5	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
26 NST	9.5	± 0.5	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
27 NST	10.5	± 0.5	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
28 NST	11	± 0.5	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
29 NST	11.5	± 0.5	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
30 NST	12	± 0.5	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
31 NST	12.5	± 0.5	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
32 NST	13	± 0.5	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)

See footnotes at end of table.

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TABLE I. Electrical characteristics - Continued.

Dash number	Attenuation dB		Maximum power input @ 25°C		VSWR maximum 1/				Weight (oz) Max	Dimension A (inches) maximum
	Nominal	Deviation DC to 18 GHz	AV (W) 3/ (continuously)	Peak 2/ (kW)	dc to 4.0 GHz	4.0 to 8.0 GHz	8.0 to 12.4 GHz	12.4 to 18.0 GHz		
33 NST	13.5	± 0.5	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
34 NST	14	± 0.5	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
35 NST	14.5	± 0.5	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
36 NST	15	± 0.5	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
37 NST	15.5	± 0.5	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
38 NST	16	± 0.7	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
39 NST	16.5	± 0.7	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
40 NST	17	± 0.7	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
41 NST	17.5	± 0.7	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
42 NST	18	± 0.7	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
43 NST	18.5	± 0.7	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
44 NST	19	± 0.7	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
45 NST	19.5	± 0.7	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
46 NST	20.5	± 0.7	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
47 NST	21	± 0.7	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
48 NST	21.5	± 0.7	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)

See footnotes at end of table.

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TABLE I. Electrical characteristics - Continued.

Dash number	Attenuation dB		Maximum power input @ 25°C		VSWR maximum 1/				Weight (oz) Max	Dimension A (inches) maximum
	Nominal	Deviation DC to 18 GHz	AV (W) 3/ (continuously)	Peak 2/ (kW)	dc to 4.0 GHz	4.0 to 8.0 GHz	8.0 to 12.4 GHz	12.4 to 18.0 GHz		
49 NST	22	± 0.7	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
50 NST	22.5	± 0.7	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
51 NST	23	± 0.7	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
52 NST	23.5	± 0.7	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
53 NST	24	± 0.7	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
54 NST	24.5	± 0.7	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
55 NST	25	± 0.7	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.5	1.240 (31.50)
56 NST	28	± 0.7	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.7	1.240 (31.50)
57 NST	32	± 1.0	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.8	1.850 (46.99)
58 NST	36	± 1.0	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.8	1.850 (46.99)
59 NST	44	± 1.0	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.8	1.850 (46.99)
60 NST	45	± 1.0	2	0.5	1.15:1	1.20:1	1.25:1	1.35:1	0.8	1.850 (46.99)

1/ VSWR value is for both ends.
2/ Peak power for a duty cycle of 5×10^{-4}; maximum pulse duration of 5 microseconds.
3/ Power input is derated linearly to 0.5 watts at +125°C.

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Referenced documents. In addition to MIL-DTL-3933, this specification sheet references the following documents:
MIL-STD-348
FED-STD-H28

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